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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MEE 323 — Past Questions & Materials
by Renance

Course: MEE 323
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 300

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Paper 1 of 3 — MEE 323 2

Level 300 • First Semester • PastQuestion

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Lecture Notes on manufacturing Process

Engineering Material**Ferrous material;**

A general classification of engineering materials is shown in Figure 1.8. Engineering material can be broadly classified as Metallic and Non-Metallic materials. The metallic materials can be further classified as ferrous and non-ferrous materials. Ferrous materials will now be discussed in the following paragraph.

Ferrous materials are those materials which contain iron as their main constituent. Other constituents are carbon, manganese, silicon, sulphur and phosphorous which exist in varying proportion to form various ferrous materials. Among these materials the following are most commonly used in engineering applications:

1. Wrought iron
2. Cast irons
3. Carbon steels
4. Stainless steels
5. Tool steels

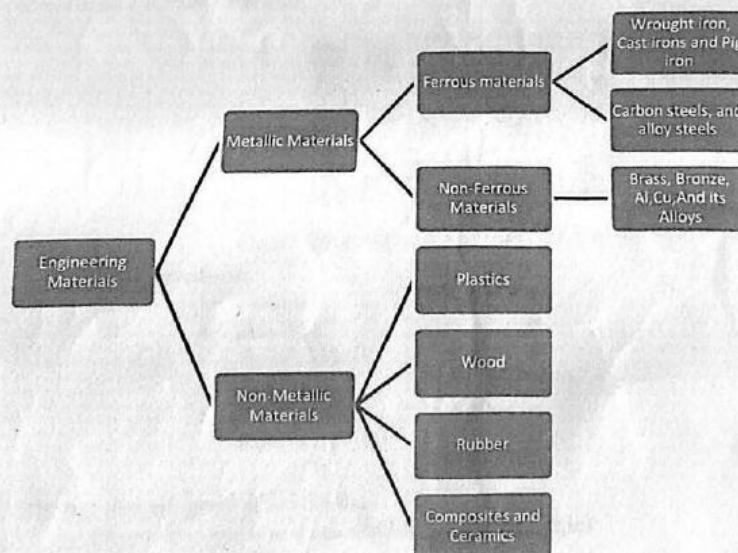


Figure Classification of materials

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Lecture Notes on manufacturing Process

Cast iron

Cast iron is an alloy of iron and carbon having a minimum of 2% to maximum of 4.5% carbon with small amount of

- Silicon (1%-3%),
- Sulphur (upto-0.15%)
- Phosphorous (upto-1%)
- Manganese (0.5%-1%)

Types of cast iron:

1. Gray Cast iron
2. White Cast iron
3. Malleable Cast iron
4. Nodular Cast iron
5. Mottled Cast iron
6. Alloy Cast iron

Gray Cast iron

In gray cast Iron the carbon (graphite) is present in the form of flakes. Due to the presence of these flakes the stress concentration at the boundaries of the flakes increases causing the material to behave as a brittle material.

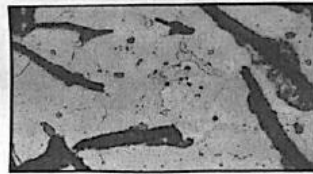


Figure Micro structure of gray cast iron at 500X

Composition:

- | | |
|-------------------|-------------|
| • Carbon- | 3%-3.5% |
| • Silicon- | 1%-2.75% |
| • Phosphorous- | 0.15%-1% |
| • Sulphur- | 0.02%-0.15% |
| • Manganese- | 0.4%-1% |
| Rest % is of Iron | |

Properties of gray Cast Iron:

1. Due to free graphite it has self lubricating properties
2. Good Machinability
3. Low tensile strength and good compressive strength
4. Brittle In Nature

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Lecture Notes on manufacturing Process

5. Good Resistance to wear

Application of Gray Cast Iron

1. Machine Tool Structures
2. pipe Fittings
3. Manhole Cover
4. piston , Flywheel
5. material Subjected to Compressive Load

White Cast Iron

Carbon presents in form of carbide (Fe_3C). White cast iron is unique in that it is the only member of the cast iron family in which carbon is present only as carbide. Due to the absence of graphite, it has a light appearance. The presence of different carbides, depending on the alloy content, makes white cast irons extremely hard and abrasion resistant but very brittle. It shows white color at fracture.

Carbide is also known as cementite and is formed by rapid cooling of molten iron.



Figure -Micro structure of white cast iron

Composition:

- Carbon—1.75%-2.3%
- Silicon— 0.85%-1.25
- Manganese— up to 0.5%
- Phosphorous—0.05%-0.2%
- Sulphur up to—0.12%

Properties of white Cast Iron

1. Extremely Hard and Brittle
2. Poor Machinability
3. High Tensile and Low Compressive Strength
4. High Melting point Than Gray Cast iron

Application of White cast Iron

In parts subjected to excessive wear or where hard coating is required i.e. outer surface of an wheel, cam, balls, brakes etc.

As raw material for malleable cast iron and wrought iron

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Malleable Cast Iron:

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Properties:

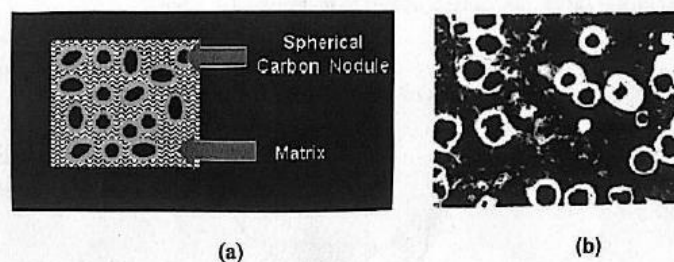
1. Excellent Machinability
2. Significant ductility
3. Good shock resistance properties

Application.

1. Transmission gears, Connecting rods and universal joint yokes,
2. Differential cases and certain gears,
3. compressor crankshafts and hubs, flanges,
4. pipe fittings and valve parts for railroad, Marine and other heavy-duty applications

Nodular Cast Iron

This structure is developed from the melt. The carbon forms into spheres when cerium, magnesium, sodium, or other elements are added to a melt of iron with very low sulfur content that will inhibit carbon from forming. The control of the heat-treating process can yield pearlitic, ferritic, martensitic matrices into which the carbon spheres are embedded.



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Lecture Notes on manufacturing Process

Figure (a) Nodular (Ductile) Cast Iron and the spherical carbon embedded into the matrix. (b) Photomicrograph of Nodular Cast iron

Properties:

1. High ductility, with grades guaranteeing more than 18% elongation (as high as 25 %), or high strength, with tensile strengths exceeding 120 Ksi.
2. Greater mechanical and wear resistance.
3. Providing tensile strengths exceeding 230 Ksi.
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Application:

Used for such critical automotive parts as crankshafts, engine connecting rods, idler arms, wheel hubs, truck axles, front wheel spindle supports, disk brake calipers, suspension system parts, power transmission yokes high temperature applications for turbo housing and manifolds, and high security valves for many applications. The cast iron pipe industry is another major user of ductile iron

Carbon steels:

Steel is an alloy of carbon and iron having carbon up to 1.5% in combined form of iron carbide.

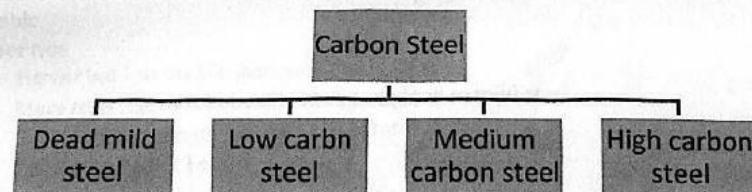


Figure- Classification of carbon steel

Dead mild steel:

- Carbon content up to 0.15%
- Use for making high ductility wire, thin sheets, wire, rod, etc.
-

Low Carbon Steel:

Also known as mild steel Contain 0.05% -0.32% carbon

Properties

1. Soft and offer good ductility
2. Tough but have low wear resistance

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Lecture Notes on manufacturing Process

3. Excellent formability
4. It is not hardened by heat treatment but in some exceptional cases carburizing and annealing can be done.

Application

Used for making screw, nuts, bolts, rivets, thin canes,, chains, small forging, wire etc.

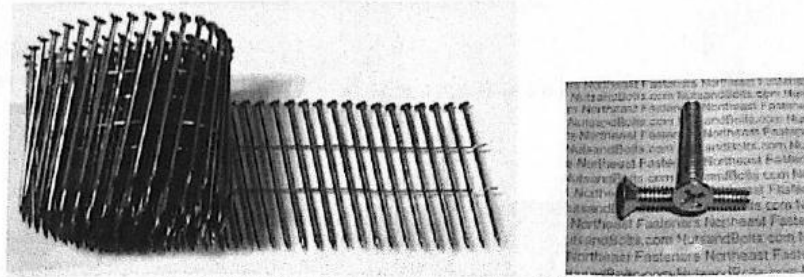


Figure Example of mild steel (low carbon steel)

Medium Carbon Steel:

Contains 0.35% - 0.5% of carbon Offer more strength and hardness but less ductile and malleable

Properties

1. Harder but less ductile than mild steel
2. More response to heat treatment as compared to mild steel.
3. Good tensile strength i. e. 600-710 N/mm²
4. Can be welded or brazed

Application

Structural steel, rails and garden tools

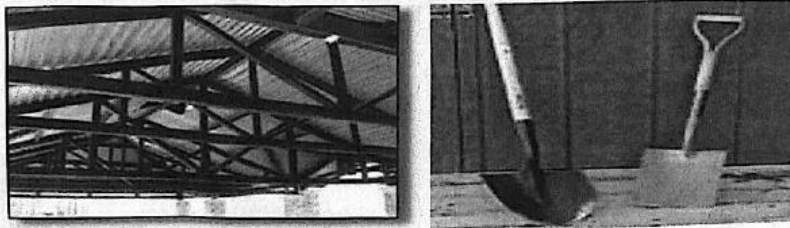


Figure 1.12 structural steel and garden tool

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High Carbon Steel

Also known as 'tool steel' Contain 0.55%-1.5% carbon

Properties

1. Highest tensile strength and hardness in plain carbon steel
2. Mechanical properties can be easily varied by various heat treatment process
3. Lowest ductility and Machinability in plain carbon steel

Application

Due to good tensile strength and hardness, high carbon steel find extensive application in hand tools, cutting tools, drill ,chisels, punches, saw blades etc.

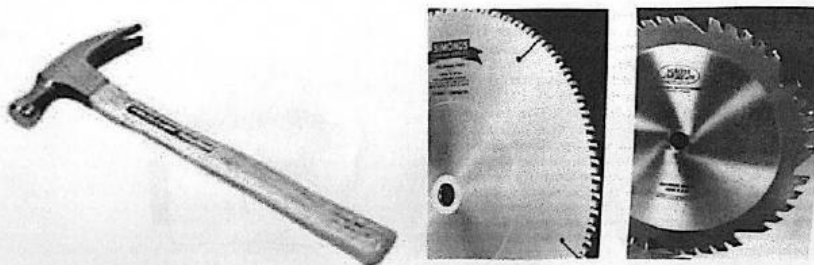


Figure- Chisel and cutter

Stainless Steel:

1. Steel alloyed with chromium (18%), nickel (8%), and magnesium (8%) Hard and tough
Corrosion resistance
2. Comes in different grades Sinks, cooking utensils, surgical instruments

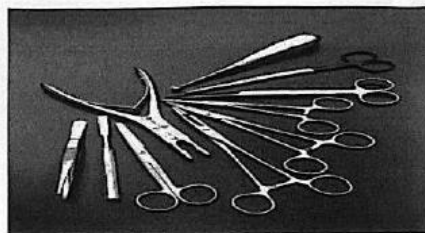


Figure -surgical instrument

Prepared by;

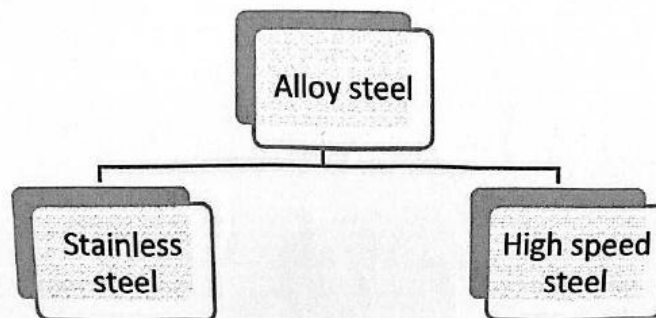
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High Speed Steel:

1. Medium Carbon steel alloyed with Tungsten, chromium, vanadium
2. Very hard
3. Resistant to frictional heat even at high temperature
4. Can only be ground
5. Machine cutting tools (lathe and milling) Drills

Alloy steel



Stainless Steel:

1. Steel alloyed with chromium (18%), nickel (8%), and magnesium (8%) Hard and tough
Corrosion resistance
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Fig- surgical instrument

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Lecture Notes on manufacturing Process

Mechanical Properties of material

Introduction: Mechanical Properties of Materials

The mechanical properties of the materials are the properties which are associated with the ability of material to resist mechanical forces and load. These mechanical properties of the materials are briefly introduced in the following paragraph.

Basic Properties of materials

Strength: It is the ability of material to resist externally applied forces without breaking of material. Strength can further be classified as

- ✓ Compressive strength
- ✓ Tensile strength
- ✓ Bending strength
- ✓ Torsional strength and
- ✓ Shear strength

Stiffness: It is the ability of material by which it can resist deformation or deflection in material under external load. This property makes the material rigid and resistant to bending.

Ductility: It is the property of material enabling it to be drawn into wire with the application of a tensile force. The ductility is usually measured in terms of, percentage elongation in length and percentage reduction in cross-sectional area

$$\% \text{ Elongation in length} = \frac{\text{change in length}}{\text{Original length}} \times 100$$

$$\% \text{ Reduction in Cross - section area(C/S)} = \frac{\text{change in C/S area}}{\text{Original C/S area}} \times 100$$

Toughness: It is the property of material to resist fracture due to high impact loads like hammer blow. It is measured by the amount of energy that a unit volume of the material has absorbed after being stressed up to the point of fracture. This property is desirable in parts subjected to shock and impact loads.

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Paper 2 of 3 — MEE 323 manufacturing technology II 2025

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN |

DEPARTMENT OF MECHANICAL ENGINEERING |

Second Semester Examination 2023/2024 Session

COURSE CODE AND TITLE: MEE 323- Manufacturing Technology rr |

DATE: 24th September, 2024 |

INSTRUCTIONS:

(i) Answer any four questions (ii) Time allowed is 2 hours |

Question 1

- What do you understand by the term forging?
- Define the following terminology used in machining study (i) cutting speed, V_c feed, and (iii) depth of cut
- Estimate the machining time to turn a M.S bar of 3 cm diameter down to 2.5 cm for a length of 10 cm in a single cut. Assume cutting speed = 30 m/min and $f = 0.4$ mm/rev.

Question 2 part a:

- Why is Wood the most widely used material for pattern making? " owe
- What are the Constituents of moulding sand? ocd
- Define die casting and state two types of die casting -
- With the aid of sketches explain the top, applicable to metal casting

Question 3 ahd

(9) Define welding and list a least four welding defects

(>) What are the disadvantages of welded joints over other joints such as riveted or bolted joints?

(c) Explain the following welding processes (i) Forge welding, and (ii) gas welding, see 7

Question 4 me eh

bottom and parting line gating system

1

- In the course of sintering process, bonding of individual particles takes place. Explain the different ways bonding take place, Give
- Explain the following generally | used powder making processes in powder metallurgy (i) atomisation, (ii) chemical reduction process, (iii) electrolytic process, (iv) sieving process and milling process | : Let us see. Let us see

eo ~ Question 5 on "eM O3t. am ae

> a. Differentiate between hot and cold working, « i

- “tes b. Applying a well labelled diagram briefly describe the Peal method of processing -
 as plastic (i) Extrusion and (ii) calendaring FeO i nai +
 c. With the aid of sketches describe the Production of plastics material from eeuge \
 oil. a BS Cash dens ey 1 le ,
 ' cheese
 uestion 6 | dese aes
 4 is re-crystallization and re-cryStallization temperature? aes
 14 y x
 b. Define heat treatment and what are the purposes of heat treatment ‘mr
 c. Describe the following heat treatment techniques: (1) bi aka and (2) 4
 Normalizing

Paper 3 of 3 — MEE 323 - Manufacturing Process Engineering Material

Level 300 • Second Semester • Handout

Lecture Notes on manufacturing Process

Engineering Material

Ferrous material;

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1. Wrought iron
2. Cast irons
3. Carbon steels
4. Stainless steels
5. Tool steels

Wrought iron,

Cast irons and Pig
iron

Ferrous materials

Carbon steels, and
alloy steels

Metallic Materials

Brass, Bronze,

Non-Ferrous

Al,Cu,And its

Materials

Alloys

Engineering

Materials Plastics

Wood

Non-Metallic

Materials

Rubber

Composites and

Ceramics

Figure Classification of materials

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Lecture Notes on manufacturing Process

Cast iron

Cast iron is an alloy of iron and carbon having a minimum of 2% to maximum of 4.5% carbon with small amount of

- Silicon (1%-3%),
- Sulphur (upto-0.15%)
- Phosphorous (upto-1%)
- Manganese (0.5%-1%)

Types of cast iron:

1. Gray Cast iron
2. White Cast iron
3. Malleable Cast iron
4. Nodular Cast iron
5. Mottled Cast iron
6. Alloy Cast iron

Gray Cast iron

In gray cast Iron the carbon (graphite) is present in the form of flakes. Due to the presence of these flakes the stress concentration at the boundaries of the flakes increases causing the material to behave as a brittle material.

Figure Micro structure of gray cast iron at 500X

Composition:

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Rest % is of Iron

Properties of gray Cast Iron:

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3. Low tensile strength and good compressive strength
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Lecture Notes on manufacturing Process

5. Good Resistance to wear

Application of Gray Cast Iron

1. Machine Tool Structures

2. pipe Fittings

3. Manhole Cover

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In parts subjected to excessive wear or where hard coating is required i.e. outer surface of a wheel, cam, balls, brakes etc.

As raw material for malleable cast iron and wrought iron

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(a) (b)

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Lecture Notes on manufacturing Process

Figure (a) Nodular (Ductile) Cast Iron and the spherical carbon embedded into the matrix. (b) Photomicrograph of Nodular Cast iron

Properties:

1. High ductility, with grades guaranteeing more than 18% elongation (as high as 25 %), or high strength, with tensile strengths exceeding 120 Ksi.
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Application:

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Carbon Steel

Dead mild Low carbn Medium High carbon

steel steel carbon steel steel

Figure- Classification of carbon steel

Dead mild steel:

- Carbon content up to 0.15%
- Use for making high ductility wire, thin sheets, wire, rod, etc.

Low Carbon Steel:

Also known as mild steel Contain 0.05% -0.32% carbon

Properties

1. Soft and offer good ductility
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Figure Example of mild steel(low carbon steel)

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Application

Structural steel, rails and garden tools

Figure 1.12 structural steel and garden tool

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High Carbon Steel

Also known as tool steel. Contain 0.55%-1.5% carbon

Properties

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Application

Due to good tensile strength and hardness, high carbon steel find extensive application in hand tools, cutting tools, drill, chisels, punches, saw blades etc.

Figure- Chisel and cutter

Stainless Steel:

1. Steel alloyed with chromium (18%), nickel (8%), and magnesium (8%) Hard and tough
- Corrosion resistance
2. Comes in different grades Sinks, cooking utensils, surgical instruments

Figure -surgical instrument

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High Speed Steel:

1. Medium Carbon steel alloyed with Tungsten, chromium, vanadium
2. Very hard
3. Resistant to frictional heat even at high temperature
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Alloy steel

Alloy steel

Stainless High speed

steel steel

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Ductility: It is the property of material enabling it to be drawn into wire with the application of a tensile force. The ductility is usually measured in terms of, percentage elongation in length and percentage reduction in cross-sectional area

change in length

% Elongation in length = $\frac{\text{change in length}}{\text{Original length}} \times 100$

Original length

change in C/S area

% Reduction in Cross – section area(C/S) = $\frac{\text{change in C/S area}}{\text{Original C/S area}} \times 100$

Original C/S area

Toughness: It is the property of material to resist fracture due to high impact loads like hammer blow. It is measured by the amount of energy that a unit volume of the material has absorbed after being stressed up to the point of fracture. This property is desirable in parts subjected to shock and impact loads.

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Lecture Notes on manufacturing Process

Figure 1.1 graphs between stress and strain showing toughness of brittle and ductile material

Elasticity: It is the property of material to regain its original shape after deformation when the external force is withdrawn. This property is desirable for materials used in tools and machines.

Plasticity: It is the property of material which allows it to permanently retain the deformation produced under load permanently. This property of material is necessary for forgings, in stamping coins and in making ornaments.

Malleability: It is a special case of ductility which permits the material to be rolled or hammered into thin sheets. A malleable material should be plastic but it is not essential to be strong enough to be drawn into wires.

Ex. Lead, soft steel, wrought iron, copper and aluminum etc.

Brittleness: It is the property of material opposite to ductility. It is the property by virtue of which a material breaks suddenly with little permanent distortion. Brittle materials when subjected to tensile loads snap off without giving any sensible elongation.

Hardness: It is a very important property of the metals and has a wide variety of meanings. It embraces many different properties such as resistance to wear scratching, deformation and Machinability etc. It imparts the ability to the metal to cut another metal.

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Lecture Notes on manufacturing Process

Fracture

A separation of an object into two or more pieces in response to active stresses far below the melting temperature of the material

Two steps in the process of fracture:

Crack initiation

Propagation of crack

Types of fracture;

1. Ductile Fracture
2. Brittle Fracture
3. Fatigue Fracture
4. Creep Fracture

Ductile fracture further classified as

(a) Highly ductile fracture

(b) Moderately ductile fracture with necking. Called a cup-and -cone fracture most common form of ductile fracture

(a) (b)

Figure 1.2 (a) Highly ductile fracture (b) Moderate ductile (cup and cone) fracture

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Ductile Fracture:

Ductile fracture is defined as the fracture or failure of material with appropriate deformation in the material. Ductile fracture takes place by slow propagation of crack after prolonged and appreciable deformation. The crack propagation Occurs at an angle of 45° , where shear stress is at a maximum, resulting in cup and cone appearance of fracture surface

Ductile fracture involves following stages

- (a) Initial necking
- (b) Cavity formation
- (c) Cavities going to form a crack
- (d) Crack propagation
- (e) Final shear

Figure 1.3 Stages in Ductile fracture (Cup and cone)

Brittle Fracture

Exhibits little or no plastic deformation and low energy absorption before failure. Crack propagation is spontaneous and rapid and occurs perpendicular to the direction of the applied stress, forming an almost flat fracture surface

Deemed unstable as it will continue to grow without the aid of additional stresses

- Under uniaxial tension loading, fracture occurs at 90 degrees with the axis of loading.
- There is no plastic deformation thus there is no necking.

The failure plane has a granular appearance.

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Figure 1.4 Brittle fracture Example of brittle fracture

Fatigue fracture:

Fatigue failure is defined as a fracture which takes place due to repeated application of fluctuating and reversing load. This fluctuating and reversing load promotes nucleation and growth of crack, which becomes microscopic in size and ultimately leads to failure. This failure occurs at stresses much below the tensile strength of the material and well within the elastic range. Tendency of fatigue failure increases with rise in temperature, high rate of straining etc. Fatigue consideration is very critical in rotating shaft axle, wing of airplane.

Figure 1.5 Reversible and fluctuating stress Example of fatigue failure

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Creep fracture

Creep is the tendency of a solid material to slowly move or deform permanently under the influence of stresses

Or

Creep is a slow permanent deformation of a metal due to steady load or prolonged loading. It is influenced by temperature so the material like copper iron and its alloys are susceptible to creep when subjected to high temperature. Some material like lead, zinc, rubber etc. also creeps at room temperature.

Ex. Gradual loosening of bolts. Sagging of long span cable etc.

Figure 1.6 Stages in Creep failure Example of creep failure

Stages in Creep failure

In the initial stage, or primary creep, the strain rate is relatively high, but slows with increasing

strain. This is due to work hardening. The strain rate eventually reaches a minimum and becomes near constant. This is due to the balance between work hardening and annealing (thermal softening). This stage is known as secondary or steady-state creep. This stage is the most understood. The characterized "creep strain rate" typically refers to the rate in this secondary stage. Stress dependence of this rate depends on the creep mechanism. In tertiary creep, the strain rate exponentially increases with stress because of necking phenomena.

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Non-Ferrous Metals and Alloys

1. Aluminum

Aluminum ore is found as a hydrated aluminum oxide, called bauxite. The impurities present in it are oxides of iron, silicon, and titanium. The process therefore, is to separate aluminum oxide from these impurities. For this purpose, bauxite is fused in electric furnace and carbon is added to reduce the impurities, which form sludge and can be removed. As a result of this refining pure aluminum oxide is separated from the impurities. Then an electrolytic bath is used to reduce aluminum from its oxide. As the electrolytic process proceeds the oxygen escapes through the bath and molten aluminum collects at the bottom, from where it is periodically tapped off.

Properties:

1. High electrical conductivity
2. High heat conductivity
3. Good resistance to corrosion
4. Highly ductile
5. Light in weight

Application

1. Used for heavy conductor and bus bar work.
2. In domestic utensils and other heat conductive appliances.
3. Used in manufacture of containers for chemical industries.
4. It can be easily worked, extruded, rolled, drawn, and forged.

2. Copper

It is not available in pure form under the earth. It is extracted from its ores through a series of processes. Copper pyrites are the main ores used for extracting copper. The copper ore is first roasted to drive out the water, CO₂ and sulphur. It is followed by melting in a reverberatory furnace of the type used for wrought iron. Silica is added to the charge to form slag with impurities like iron and alumina, etc. The molten metal is tapped and transferred to a convertor where air is blown through it to burn the impurities. This result in the production of crude form of copper, known as blister copper, containing 68% purity. Final refining is done by an

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electrolytic process, pure copper depositing on the cathode. This gives a highly pure (99.9%) copper which is re melted and cast into suitable shapes.

Properties:

1. High electrical conductivity.
2. High heat conductivity
3. Good corrosion resistance
4. High ductility
5. Light in weight

Application:

1. Used as electrical conductor in various shapes and forms viz., sheet and contacts.
2. Used in heat exchanger and heating vessel and appliances.
3. Used for providing base coating on steel prior to nickel and chromium plating.
4. Can be easily cold worked, rolled, and drawn.
5. Used in various appliances where light weight is desired.

3. Magnesium

Principle sources for obtaining magnesium are natural salt brines, sea water, and water liquors obtained from potash industry and ores. The principle ores are magnesite, dolomite and carnallite. Various processes have been developed for its extraction, but the most popular and widely used one is electrolytic process

Properties

1. Lightest of all metals, weighing about two-third of aluminum.
2. It may be sand, gravity and pressure die-cast.
3. Its casting is pressure tight and obtains good surface finish.
4. It may be easily formed, drawn, forged and machined with high accuracy.
5. Addition of 10% aluminum and small amount of zinc and manganese improve its strength and casting characteristic.
6. Addition of 2% Mn helps in its easy forming into plates and sheets and extrusion works.

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Application: Motor car gear box, differential housing, and portable tools

4. Zinc

The zinc ore is first concentrated through suitable process. This concentrate is fed into a retort with suitable amount of carbonaceous material (say coal). Several retorts are housed in one furnace and their temperature raised to 1100°C. Zinc emerges as a vapour and is passed through a condenser. By rapid cooling the zinc vapour may be quickly converted into powdered zinc.

Properties

1. High corrosion resistance
2. Low melting point
3. High fluidity

Application: Widely used as a protective coating on iron and steel. Low melting point and high fluidity make it the most suitable metal for pressure die-casting.

5. Lead

Lead ores are generally found as oxides or sulphides. Other impurities present in the ores are iron, copper and zinc etc. The prepared ore concentrate, together with the flux (lime and silica), is fed into a small blast furnace, where the temperature is raised to about 1010°C. The lead is melted and a liquid slag formed of the impurities. Both slag and molten lead are tapped at intervals. Further refining is carried out in a reverberatory furnace where an oxidizing atmosphere is maintained to burn out the impurities.

Properties:

1. Good corrosion resistance
2. Good resistance to chemical action.
3. It is soft heavy and malleable
4. It is alloyed with brass and steel to impart them free cutting properties.

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Application

1. Used for water pipe and roof protection
2. Used for acid bath and containers in chemical industry.
3. Can be easily worked and shaped.
4. Used as an alloying element in making soft solders and plumber's solders.

6. Tin

The most prominent tin ore is cassiterites. It also carried compounds of copper, iron, lead antimony, bismuth and zinc etc. As usual an ore concentrate is prepared. This concentrate is roasted to drive off excess arsenic and sulphur. The roasted ore is transferred to a reverberatory furnace, where it is heated. Anthracite is added to the charge which reacts chemically to separate tin, the latter sinking to the bottom of the furnace. From there it is tapped at intervals. This crude

tin is re melted and re find further.

Properties

1. Good resistance to acid corrosion
2. It is soft, has good plasticity and can be easily worked.
3. It can be easily rolled in thin foils, but cannot be drawn due to low strength..

Application

It is used as an alloying element in soft solders, bronzes bearing metals.

7. Nickel

Its extraction process consists of first roasting the ore, followed by melting in a small blast furnace. Lime stone and quartz are added as flux. They form slag with impurities. Coke is used as fuel. Crude molten nickel tapped off periodically from the bottom of the furnace. This crude metal is further refined in a Bessemer convertor followed by treating with sulphuric acid to extract pure nickel. Copper is separated as a copper sulphate.

Properties

1. It has good resistance to both acid and alkali corrosion

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2. It has high tensile strength and can be easily worked cold and hot.

Application

1. It is widely used in food processing.
2. It is plated on steel to provide a corrosion resistant surface
3. It is important alloying element with steel.

Non-ferrous alloys

Due to poor physical and mechanical properties and high costs the non-ferrous metals are seldom used in their in pure state. But their alloys carry very good physical and mechanical properties and are widely used. In general they have lower strength, hardness, and modulus of elasticity than irons and steels. However they carry their own importance due some exceptional properties possess, like lightness, ease in fabrication, good machinability, high resistance to corrosion, attractive appearance and good cast ability etc. Some commonly used non-ferrous alloys are described in the following articles:

1. Brasses

All brasses are basically alloys of copper and zinc. There are two main varieties of brasses:

1. Alpha brass (Zn up to 37%) - for cold working.
2. Alpha beta brass (Zn 33% to 46%) - for hot working.

Alpha brasses are very ductile and can be readily cold worked without any chances of fracture.

They can be cold rolled into sheet, drawn into wire, deep drawn and drawn into tubes. In these brasses, as the proportion of zinc increases, their strength increases but ductility decreases. They are worked hardened when subjected to intensive cold working, but ductility can be regained by annealing them at 6000 C. Slow cooling provides maximum ductility, but for common uses they may be water quenched. Deep drawing of this brass requires periodical annealing during the process.

Alpha beta brasses lose strength at high temperature but become very plastic. It, therefore respond very well to hot rolling, hot extrusion, hot stamping and casting, etc. when cold worked, fracture are likely to develop.

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Common types of brasses in engineering use are the following:

- a) Cartridge brass: - It has 70% Cu and 30% Zn. It is very strong and ductile. It is used for wide range of drawn components like cartridge cases, head lamp reflector, radiator shells and drawn tubes.
- b) Muntz metals: - It contains 60% Cu and 40% Zn and can be cast, rolled, extruded and stamped. It is a sort of general purpose alloy having good resistance to corrosion. It is used for casting pump parts, valve, taps and other similar items.
- c) Naval brass: - It contains 60% Cu, 39% Zn and 1% tin. It is more or less similar in composition to Muntz metals except that 1% Zn is replace by 1% tin. As a result of this change the resistance to sea water corrosion is vastly improved. This alloy is therefore, widely used for cast and forged fittings for ships.
- d) Admiralty brass: - It contains 70% Cu, 29% Zn and 1% tin. It is similar to cartridge brass in composition except that 1% Zn is replaced by 1% tin. It can be cold worked and has good resistance to sea water corrosion. It is cold drawn into tubes and rolled into sheets and bars. It is widely used in ship fitting, bolts, nuts, washer and other items subjected to sea water corrosion.
- e) Gilding brass: - it contain up to 15% zinc and the rest Cu. It is very cold working alloy and is used for jewelry, decorative and ornamental work. It is commercially available as cold rolled strip, wire or sheet. Its color according to the percentage of Zn varies from red to bright yellow. It is also called Gilding metal.
- f) Delta brass: - It is also known as delta metal, it consists of 60% of Cu, 37% Zn and 3% iron. It can be easily hot worked, forged, rolled, extruded and cast. It has fairly good tensile strength after hot working and casting. It also has a good corrosion resistance. It can suitably replace steel castings.
- g) Free cutting brass: - It contains 60% Cu, 37%Zn and 3% Pb. It is specially used in machining work, such as producing components from bar stock on turret and automatic

lathe. It is also used for making cast, forged, or stamped, blanks to be used for further machining. With this metal very high speeds and feeds can be employed in machining.

h) Beta brass: - It contains 50% Cu, and 50% Zn. Higher percentage of zinc render it hard and brittle, but it softens quickly when heated and melts at 8700 C. Its main application is brazing solder.

i) Silicon brass: - It contains 80% Cu, 16% Zn and 4% Si. It responds well to welding and is widely used for refrigerators and fire extinguisher shells. It can also be easily sand or

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gravity die cast, hot stamped and extruded. It can be used as a cheaper substitute for phosphor bronze.

2. Bronzes

Bronze is basically an alloy of copper and tin. In general, it possesses superior mechanical properties and corrosion resistance than brass. Those containing up to 8% tin are called working bronze. They can be easily cold worked, rolled, formed and drawn. They are available in various forms, as strip, wire and sheet etc.

With the increase in tin content, its strength and corrosion resistance increase. It is then known as hot working bronzes. Small addition of phosphorous further improves its strength, ductility and bearing properties. The amount of phosphorous added is 0.5%. This is then known as phosphorous bronze.

a. Phosphorous bronze: - Widely used in the several manufacturing units, we present an array of phosphor bronze alloys rods, phosphor bronze bars, phosphor bronze squares and many more. These phosphor bronze alloys offer resistance towards fatigue and chemical corrosion, so they are used for varied application in various industries such as in heavy engineering, defense, aviation, electrical and more.

b. Gun metal: - The term 'gunmetal' is applied to a family of copper-based casting alloys containing between 2-11% tin and 1-10% zinc. Modified forms may contain, in addition, such elements as lead (up to 7%) and nickel (up to 6%) when the alloys are classified as 'leaded gunmetal' and 'nickel gunmetal'. Gunmetal is noted for the manufacture of intricate castings required to be pressure tight such as valves, pipe fittings and pumps.

c. Aluminum bronze: - Aluminum bronzes are used for their combination of high strength, excellent corrosion and wear resistance. Aluminum bronze alloys typically contain 9-12% aluminum and up to 6% iron and nickel. Alloys in these composition limits are hardened by a combination of solid solution strengthening, cold work, and precipitation of an iron rich phase. High aluminum alloys are quenched and tempered. Aluminum bronzes are used in marine hardware, shafts and pump and valve components for handling seawater, sour mine

waters, and industrial process fluids. They are also used in applications such as heavy duty sleeve bearings, and machine tool ways.

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d. Silicon bronze: - it contains 1 to 4% Si, 0.25% to 1.25% Mn, 0.5% to 1% iron (if added) and the rest copper. Small addition of upto 0.5% Pb will improve machinability. It has high strength, toughness, and corrosion resistance. It can be readily hot worked. With low silicon content it can be safely cold worked also. It is widely used for boiler parts, tanks marine hardware and similar other item.

e. Manganese bronze: - It contains 55% to 60% copper, 38% to 42% Zn, upto 1.5% tin, upto 2% iron,

upto 1.5% Al and upto 3.5% Mn. It has superior mechanical properties and high corrosion resistance. It has poor respond to cold working but can be readily hot worked. It is used for such part

where high strength and corrosion resistance are desired, such as in ship propellers and rudders, etc.

Aluminum alloys

a. Duralumin: - It contains 4% Cu, 0.5% Mg, 0.5% Mn and the rest aluminum. Duralumin alloys are relatively soft, ductile, and workable in the normal state; they may be rolled, forged, extruded, or drawn into a variety of shapes and products. Their light weight and consequent high strength per unit weight compared to steel suit them for aircraft construction. Because duralumin loses strength in welding, a special laminated sheet form called alclad is used for aircraft construction; it has thin surface layers of pure aluminum or a corrosion-resistant aluminum alloy covering the strong duralumin core.

b. Aluminum casting alloy: - It contains 90% Al, 8% Cu, 1% iron and 1% Si. It has good strength, hardness and machinability. It may be sand, pressure or gravity die cast.

c. Y-alloy: - It contains 93% Al, 4% Cu, 2% Ni, and 1% Mg. Its principle use is as a casting alloy. It maintains its strength at elevated temperatures, and is used for pistons of I.C. engines. A treatment of Y-alloy castings, consisting of quenching in boiling water from a temperature of 5100 C and then aging for 5 days, develops very good mechanical properties in them. It is also used in strip and sheet forms.

Nickel alloys

a. German silver: - The name German silver or nickel silver is misleading since it has no silver content, but is an alloy of 60% Cu, 20% Ni and 20% Zn. It resembles silver but instead of the soft white color it has a yellow-grey tinge.

German silver is used extensively in the hollow-ware and flatware industries as E.P.N.S. (electro-plated nickel silver), which is a nickel silver item with a plating of silver. Nickel

silver is slightly more difficult to work for jewelry purpose, than copper and other metals, since it is a harder alloy. Again it is prone to tarnish on exposure to air and will turn the skin green with prolonged contact.

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b. Constantan: - It contains 45% Ni, 55% Cu. It has high specific resistance, which is unaffected by temperature variation. It is used for accurate resistors, thermocouples, Wheatstone bridge, low temperature heater and resistances.

c. Monel metal: - It contains 68% Ni, 30% Cu, 1% iron and small additions of Mn and other elements. It is a nickel-copper alloy with high strength and excellent corrosion resistance in a range of media including sea water, hydrofluoric acid, sulfuric acid, and alkalis. Used for marine engineering, chemical and hydrocarbon processing equipment, valves, pumps, shafts, fittings, fasteners, and heat exchangers.

d. Inconel: - It contains 80% Ni, 14% Cr and 6% iron. Inconel Alloys Are Generally Known For Their Resistance To Oxidation And Their Ability To Maintain Their Structural Integrity In High Temperature Atmospheres. There Are Several Inconel Alloys That Are Used In Applications That Require A Material That Does Not Easily Succumb To Caustic Corrosion, Corrosion Caused Purity Water, And Stress-Corrosion Cracking. While Each Variation Of Inconel Has Unique Traits that make it effective in different circumstances, the majority of the alloys are used frequently in the chemical industry.

e. Nichrome: - Like Inconel it is also a nickel chromium alloy which is extensively used in electrical appliances as resistance wire.

f. Incoloy: - It is also a Ni based alloy which is widely used as a high temperature alloy. It consists of 42% Ni, 13% Cr, 6% Mo, 2.4% Ti, 0.04% C and rest iron.

g. K-Monel: - It possesses similar composition as monel, but about 3 to 4% Al is added to it. It carries similar application as monel, but has better mechanical properties than that.

h. Nimonic alloy: - It contains 80% Ni and 20% Cr. It has high strength and ability to operate under intermittent heating and cooling conditions. It is widely used in gas turbine engines.

Bearing metals

Bearing metal should possess the following important characteristics:

1. It should have enough compressive strength to possess adequate load carrying capacity.
2. It should have good plasticity to allow for small variation in alignment and fitting.
3. It should have low co-efficient of friction to avoid excessive heating.

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Some important bearing metals are the following:

- a. Babbit metal: - It is white metal containing 85% tin, 10% Sb and 5%Cu. It is used for heavy duty vehicles.
- b. Lead alloy: - It contains 40% lead and 60% cu. It may be cast in position or fused as a thin shell to a bronze or steel reinforcing shell outside.
- c. Phosphor bronze: - With 10% tin is used for light load low speed bearings. It can be sand and centrifugal cast.

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Metal Casting Processes and Their Applications

Introduction to Foundry Processes

We can shape metal into useful products by several methods; and casting offers the minimum distance between the raw material and the finished product. Casting processes basically consist of pouring a molten metal in a mould cavity where upon solidification the metals assumes the shape of the mould cavity. This product is called a casting.

Sand casting

The important terms used in metal casting are briefly explained below;

1. Pouring basin: A small funnel shaped cavity at the top of the mold into which the molten metal is poured.
2. Sprue: The passage through which the molten metal, from the pouring basin, reaches the mold cavity. In many cases it controls the flow of metal into the mold.
3. Gate: A channel through which the molten metal enters the mold cavity.
4. Riser: A column of molten metal placed in the mold to feed the castings as it shrinks and solidification. This is also known as "feeder head".
5. Runner: The channel through which the molten metal is carried from the sprue to the gate.
6. Pattern: It is the replica of the final object to be made. The mold cavity is made with the help of a pattern
7. Flask: A metal or wood frame, without fixed top or bottom, in which the mold is formed.
8. Parting Line: This is the dividing line between the two molding flasks that makes up the mold. It divides the mould into two halves or it is so chosen that more volume of casting goes

to the drag.

9. Moulding Sand: Sand, which binds strongly without losing its permeability to air or gases. It is a mixture of silica sand, clay and moisture in appropriate proportions.

10. Core: A separate part of the mold made of sand mixed with organic binders and generally baked to develop strength. These core are used to create holes, openings and various shaped cavities in the castings

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Figure-Cross-sectional view of sand mold

Steps in Making Sand Castings

There are six basic steps in making sand castings:

1. Patternmaking
2. Core making
3. Molding
4. Melting and pouring
5. Cleaning

Pattern and allowance

Pattern Material

1. Wood: Wood is the most widely used material for pattern making because it has following the properties,

1. It is easily available
2. It has low weight.
3. It is easy to work with wood.
4. It is relatively cheap.

2. Metals and alloys: Metals used for pattern making are cast iron, aluminum and copper alloy. Choice of metals depends widely upon quality and quantity of mould to be

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produced. For large scale production cast iron is used as it is cheap, easy to machine and wear resistant. Aluminum alloys are used for small scale production as it has lesser wear resistance.

Types of Pattern

1. Single Piece Pattern or solid pattern: solid pattern is made in single piece and is used when shape is relatively simple and does not create withdrawal problem, it is inexpensive and made from wood
2. Split Pattern or Two Piece Pattern: Some pattern cannot be made in single piece due to difficulty in removing pattern from mold. Pattern is split into two parts by parting lines cope and drag. Tapered pins in the cope with holes in drag for precise positioning.
3. Gated Pattern: This is an improvement over the simple pattern where the gating and runner system is integral with the pattern. This would eliminate the hand cutting of the runners and gates and help in improving the productivity of a molder
4. Loose Piece Pattern: Loose piece patterns are used when some irregularity is present on the contour of the pattern, making a difficulty in withdrawing the pattern from the mould. So to overcome this difficulty the irregular part of the pattern is held loose against the contour of the pattern using dowel pins or wire. By using such arrangement during the withdrawal we can first withdraw the main pattern which is regular in shape and then the loose pieces can be removed without damaging the contour of the casting from the gap provided by the main pattern.
5. Match plate: Provide a strong and a rigid mounting for pattern and help in eliminating the mismatch between the cope and drag half of the mould during the assembly of machine moulds. Usually all high production work on moulding machines uses the match plate pattern.
6. Follow board pattern: Used with single or multiple gated pattern. Pattern requiring follow boards are difficult to make as split pattern. The pattern rests in the follow board up to the parting line. In this position, it serves as a moulding board for the first moulding operation.
7. Sweep Pattern: many moulds of regular cross-section are made by the use of sweep patterns as shown in figure. It eliminates expensive pattern construction.

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(a) Solid (b) Split

(c) Gated (d) loose piece

(e) Match plate (f) Follow board

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(i) (ii)

(g) Sweep patterns

Figure-Types of patterns

Pattern allowances

Dimensions of pattern differ from the final casting. This difference of dimension is called allowances and is provided to ensure easy withdrawal of pattern and get dimensionally correct finished casting. There are mainly five types of allowances provided on a pattern.

1. Shrinkage allowance: when metal cools from pouring temperature to room temperature, it contract. This contraction takes place in three stages-

- a. Contraction of liquid from pouring temperature to freezing temperature (liquid contraction).
- b. Contraction due to Change of phase from liquid to solid known as solidification contraction (solid contraction)
- c. Contraction of casting from freezing temperature to room temperature known as solid contraction (solid to solid contraction)

First two contractions are compensated by riser and gates while last contraction is considered for shrinkage allowance. Stages of shrinkage are shown in given figure.

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Figure- Stages of shrinkage

2. Draft Allowance: The draft allowance is defined as the tapering which is provided on the pattern. When the sides of the pattern are deeper, it may damage the walls of the pattern while with drawing it from the mould cavity. To overcome this problem a taper or slight inclined surface is furnished on the sides of the pattern for the easy withdrawal of pattern from the mould. Table below shows the values of taper to be provided for easy removal.

(a) (b)

Figure (a) Draft and (b) Distortion allowance

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3. Distortion or camber allowance: This is applicable to casting of irregular shape such as

$<V=$, $<U=$ and $<T=$ section. When such castings are allowed to cool they are distorted due to non-uniform contraction. To eliminate this, the shape of pattern should be given allowance of equal amount in opposite direction.

4. Machining allowance: when an object is prepared using casting, the surface finish we get is very bad and rough. To overcome this issue some extra metal is provided on the surface to machine and make the surface smooth. This allowance is known as machining allowance.

Figure -Machining Allowance

5. Shake or Rapping allowance; When a pattern is to be drawn from the mould, it is first rapped or shaken, by striking over it from side to side, so that its surface may be free of the adjoining sand wall of the mould. As a result of this the size of the mould cavity increases a little and a negative allowance is to be provided in the pattern to compensate the same. However, it may be considered negligible for all practical purposes in small and medium sized casting.

Desirable properties of moulding sand

1. Inexpensive in bulk quantity
2. Retain properties through transportation and storage
3. Uniformly fills a flask or container
4. Compacted or set by simple methods
5. Sufficient elasticity to remain undamaged during pattern withdrawal
6. Withstand high temperatures and maintain its dimensions until metal solidifies Sufficient permeable to allow the escape of gases
7. Sufficiently dense to prevent metal penetration
8. Sufficiently cohesive to prevent washout of mold

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9. Chemically inert to metal being cast
10. Good collapsibility to permit easy removal and separation of casting
11. Can be recycled

Constituent of moulding sand

12. Silica (SiO_2) = 86%-90%
13. Aluminum (Al_2O_3) = 4%-8%
14. Iron oxide (Fe_2O_3) = 2%-5%
15. Clay as binder
16. Moisture to provide elasticity

Gating system

Gating system shown in figure provides a passage for the molten metal to enter the mould cavity,

fill it completely and rise in the riser. The system is composed of:

1. Pouring cup 4. sprue
2. Runner 5. gates
3. Riser

Figure -Component of gating system

Basic Requirement of a gating system

1. A gating system should avoid sudden or right angle changes in direction as it causes:

- Mould erosion
- Turbulence
- Gas pick-up

2. Fill the mold cavity completely before freezing

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3. Help to promote temperature gradients favorable for proper directional solidification
4. Should incorporate traps for separation of non-metallic inclusions(slag, broken sand, etc)
5. Should regulate proper rate at which metal enters into the mould cavity.
6. Be practical and economical to make.
7. The system should consume least amount of metal

Type of Gates

1. Top gating system: - In this type of gate the entire molten metal enters the mould cavity at the top, with the result the hottest metal remains at the top of the casting and therefore proper temperature gradient is formed to enable directional solidification of casting from the bottom towards the riser. The gate itself serves as the riser.

Drawback:-

1. Erosion of mould takes place.
2. Turbulence occur in gate
3. Excessive absorption of gases takes place
4. Not favored for non-ferrous castings

(a) (b)

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(c)

Figure (a) Top gating (b) Bottom gating and (c) Parting line gating system

2. Bottom gating system: - In this type of gate the molten metal enters the mould cavity at or near its bottom. It provides a smooth flow of molten metal reduces considerably the erosion of mould and core and minimize turbulence. Inside the cavity the metal rises gently upwards.

Drawback:-

a. In large and deep casting, the metal cools gradually as it rises up. With the result an unfavorable temperature gradient is produced i.e. the hottest metal remaining at the bottom and coolest metal at the top.

b. The directional solidification adversely affected.

3. Parting line gating system: - These gates are the simplest both in nature and construction. In this type of a gate the metal enters the mould cavity at the parting line. Such gates are very commonly employed and are found to give very satisfactory service except when the mould is very deep.

Drawback

1. It is not suitable for deep mould as the metal has to fall on the mould bottom through a considerable distance between the gate and the bottom.

2. Dropping of metal from such height cause erosion or washing of mould.

3. In non-ferrous metal this drop promotes dross and air absorption by metal, leading to an inferior casting.

Classification of Casting Defects

A large number of defects occur in sand castings produced through various methods. The major factor responsible for these defects includes:

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1. Design of casting 4. Mould and core system design

2. Pattern design 5. Gating and feeder head design

3. Mould and core materials used 6. Melting and pouring

7. Cast metal metallurgical factors

Some of the most common defects are discussed below with their causes and appearances.

The defects are discussed in the order of their importance.

Surface Defects:

1. Wash; It is low projection on the drag face of a casting that occurs due to sand being

washed away due to improper molten metal flow during pouring. The sand that is washed away appears at some other location as sand inclusion. Figure (a)

2. Rat tails or buckles: Slight compression failure of a thin layer of moulding sand causes rat tails, a more severe compression failure causes buckles (due to buckling of sand). Figure (b).

3. Swell: Swell occurs due to enlargement of mould cavity due to ferrostatic pressure. Figure (c)

4. Misrun: This defect occurs due to incomplete mould filling due to low pouring temperature, inadequate metal supply, and improper gating design. This defect determines the minimum thickness that can be cast for a given metal, superheat and type of mould. Figure (d)

(a) (b) (c) (d)

(e) (f) (g) (h)

Figure -various casting defects appear on surface

5. Hot tear and shrinkage: Hot tears are ragged edge cracks that occur due to tensile stresses during solidification. They occur due to excessive mould hardness

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discontinuity in the metal casting due to hindered contraction and occurs just after the metal has solidified. It is shown in figure (e) and (f). Improper metallurgical and pouring temperature control causes this defect.

6. Shift/Mismatch: It is due to shift of individual parts of casting due to wrong assembly of cope and drag of mould cavity, with respect to each other. Figure (g) and (h)

Internal Defects Blow holes, Porosity, Pin holes, Inclusions, Dross etc

1. Blow Holes: Blow holes appear as cavities in a casting. They may appear at the cast surface as open blows and when they occur below the surface they are called simply as blows. They occur due to entrapped gases in the metal when vent holes are not properly provided or sand porosity is poor. Figure (a).

(a) (b) (c) (d)

Figure -various casting defects (internal)

2. Porosity: This porosity occurs due to gases mainly hydrogen absorbed by molten metal which are released on cooling. Effect of micro porosity on metal ductility is shown in figure (b). Porosity may also be caused by shrinkage.

3. Cold shut: A cold shut is an interface within casting that forms when two metal streams meet without complete fusion. This defect occurs when the metal streams meet at

temperatures low enough to cause fusion

Die Casting:

Die casting is a precision casting technique in which liquid metal is injected into metallic die under high pressure. It is used to for production of fine surface, complex shapes and narrow sections. Die casting is done by

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1. Hot chamber die casting: There is a pump consisting of plunger and cylinder both made of refractory material. The liquid metal fills the refractory cylinder through the hole. When plunger moves down the metal in the cylinder is forced into the mould cavity through the goose neck. For zinc castings the pumping pressure is up to 40 MPa.

Figure (a) Hot Chamber dies casting (b) Cold chamber die casting

2. Cold chamber die casting: A measured quantity of liquid metal is poured in refractory cylinder. Plunger moves and pushes this liquid metal into mould cavity. Pressure exceeding 40 MPa are used for magnesium and aluminum alloys

Cupola Furnace

The majority of gray and white cast iron are melted in cupola furnace. As shown in figure 4.11, the cupola essentially is a vertical cylindrical steel shell with a refractory lining, supported on heavy steel or iron legs so that its lower end 3 to 5 feet above the floor. The lower end is closed by hinged doors that can be opened to permit cleaning the inside. A sand bottom, 4 to 6 inches deep and sloping downward to the pouring spout, is prepared on top of the closed bottom doors. The pouring spout has taphole 1 to 2 inches in diameter.

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Figure-Cupola furnace

After the sand bottom has been prepared, kindling wood is placed on the bottom and a bed of coke on top of this. The wood is ignited and the coke bed is increased until it is 18 to 48 inches in depth, depending on the size of the cupola. Natural draft is supplied through the tuyeres, which are openings around the periphery of the cupola about 15 to 20 inches above the top of the sand bottom. When the coke bed has been built up to the desired depth and is thoroughly ignited, alternate charges of coke and iron are added through the charging door located about half the

distance up the height of the cupola. The metal charge may be either pig iron or scrap. About one part of coke, by weight, is used to eight or ten parts of iron. In addition, small amounts of limestone, of other materials; may be added to act as a flux and to increase the fluidity of the metal. It is possible to add certain alloy-rich materials to alter the chemical composition of the final product, although such additions often are added in the ladle at the time the cupola is tapped.

After the iron and coke charges have been built up to the level of the charging door, the charge is heated for about half an hour under natural draft. The natural draft openings then are closed, and the lower is started. Under forced draft, the temperature in the cupola rise rapidly, and the iron charge start melt and run down through the coke. Final melting occurs in the zone just above the tuyeres. Molten metal should start to appear within ten minutes after the air blast is turn on. Quartz- covered peepholes make it possible to see the metal flowing down through the incandescent coke. As soon as the first metal is starting to flow, the taphole is plugged with a conical- shaped plug of fire clay.

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The molten metal collects in the lower portion of the cupola between the sand bottom and the slag spout, located slightly below the tuyeres. Excess slag floats on top of the molten metal and can be drained off through the slag hole to obtain cleaner metal. When a sufficient amount of molten metal has collected, the air blast is turned off temporarily, the clay plug is punched out, and the metal is permitted to flow out into a pouring or holding ladle. When sufficient metal has been removed, the taphole is again plugged and the air blast is turned on.

From this point the operation of the cupola is a semi continuous batch-type process. As metal is melted and removed, additional charges of coke and iron are added through the charging door. It is desirable to maintain the charge up to the level of the charging door at all times to obtain uniform combustion and melting condition. The tapping process may be continuing upto as long as fourteen to sixteen hours.

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Heat Treatment

Elementary Introduction

A process used to alter the properties and characteristics of metals by heating and cooling.

Cold working □ induce stress in metal □ lead to work hardening □ prevent further work from taking place

Three stages of heat treatment;

1. Heat the metal to the correct temperature
2. Keep it at that temperature for a the required length of time (soaking)
3. Cool it in the correct way to give the desired properties

Classification of heat treatment process;

The heat treatment processes can be classified as shown in figure

Full

annealing

Process

annealing

Annealing

Spheriodize

annealing

Normalizing

Diffusion

annealing

Heat

Hardening

treatment

Tempering Carburizing

Case Pack

Nitriding

hardeni ng carburiz ing

Liquid

Carburizing

carburizi ng

Gas

carburizing

Figure Classification of heat treatment process

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Purpose of heat treatment:

1. To reduce residual stress set up during cold working.

2. To remove gases trapped during casting and welding.
3. To refine grain structure
4. To improve mechanical properties

1. Annealing

Annealing is the process whereby heat is introduced to mobilize the atoms and relieve the metal from internal stresses. After annealing, it allows the metal to be further shaped. It involves the recrystallization of the distorted metal structure.

Annealing further be classified as shown in figure

Annealing

Full Annealing (30 Process Spheroidize Diffusion

Annealing (below Annealing (above annealing (Temp

$^{\circ}\text{C}$ -50 $^{\circ}\text{C}$ above - 1100 $^{\circ}\text{C}$ -1200

the lower critical e LCT (730 $^{\circ}\text{C}$ -

higher critical)

550 $^{\circ}\text{C}$ -650 $^{\circ}\text{C}$) 770 $^{\circ}\text{C}$) $^{\circ}\text{C}$)

Figure- Classification of annealing process

a) Full Annealing:

Main objective of full annealing are to refine the grain structure, Soften the metal, relieve the metal of internal stresses,

Heating temperature—

30 $^{\circ}\text{C}$ -50 $^{\circ}\text{C}$ above higher critical temperature for hypo eutectoid steels.

30 $^{\circ}\text{C}$ -50 $^{\circ}\text{C}$ above lower critical temperature for hyper-eutectoid steel and slowly cooling the part to be fully annealed in furnace

It is also known as higher temperature annealing process which removes all structural imperfection by complete recrystallization. In full annealing process steel is heated to a specified

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temperature after heating to that temperature metal is hold at that temperature to sufficient time to allow internal changes in its crystal structure. The metal is now slowly cooled in the furnace.

Figure- Form of steel on the basis of % c and temperature

b) Process Annealing;

The main objectives of process annealing are:

1. To remove ill effect of cold working.
2. To soften the metal so that its ductility can be further restored for plastic deformation

Heating temperature:

Slightly below the lower critical temperature

Generally- 550°C - 650°C

Holding time 2-4hr depending upon the metal thickness and then cooling in furnace or air to soften the metal

Process annealing increases-

- Ductility and plasticity
- reduces hardness, removes internal stresses
- and improves Machinability,

This Process is used in wire and sheet industries.

c) Spheroidize Annealing:

Any methods through which spheroids are formed are called as spheroidized annealing. If an annealed product contains globules of cementite (Fe_3C) in the matrix of ferrite (Fe_2O_3) in the microstructure then it is termed as a spheroid

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Heating-slightly above LCT (730°C - 770°C)

(a) (b)

Figure- (a) Globules of cementite (b) spheroidizing process

d) Diffusion annealing:

□ Also known as homogenizing annealing.□

□ Remove heterogeneity in chemical composition of heavy castings.□

□ Heating Temp- 1100°C - 1200°C at which diffusion occurs and austenite grain are homogenized□

□ Holding time is very short.□

Cooled in furnace for 6-8 hours to 800°C and then further cooled in air

Normalizing

□ This process is only confined to steel. It is used to refine the grains due to work hardening□

□ It involves the heating of the steel to just above its upper critical point(45°C - 50°C)□

□ Holding time-15 minutes□

□ Cooling at room temperature in still air□

Tempering, Quenching and hardening

Hardened steel is heated to temperature below the LCT to transform the hard and brittle martensite into ferrite and cementite.

In this process ductility and toughness achieved at the cost of hardness and strength. Tempering processes is further classified as

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Low temperature tempering (150°C - 250°C):- Relieve internal stresses

Medium temperature tempering (350°C - 450°C):- Ductility and toughness achieved

High temperature tempering (500°C - 650°C):- Complete elimination of internal stress

Quenching is defined as a process of sudden cooling of metal. When heated metal from furnace is suddenly dipped into oil or water then this process is called quenching. Due to quenching sudden contraction takes place due to sudden heat loss from the metal. Sudden contraction caused crack in metal.

Hardening

Hardening process consist of heating the steel to a temperature above its critical point, holding it at that temperature for sufficient time depending upon its thickness and then quenching in suitable medium such as water, oil or molten salt bath

1.3.5 Case hardening

Classification of case hardening processes is shown in figure 1.20

Cyniding

case pack

Nitriding

Hardening carburizing

liquid

Carburizing

carburizing

gas

carburizing

Figure -Classification of Case hardening

Case hardening produces a hardened case or surface over a tough, shock resistant core. Case hardening is done by the following method

1. Carburizing

Carburizing is a process of adding Carbon to the surface. This is done by exposing the part to a Carbon rich atmosphere at an elevated temperature and allows diffusion to transfer the Carbon atoms into steel. This diffusion will work only if the steel has low carbon content, because

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diffusion works on the differential of concentration principle. If, for example the steel had high carbon content to begin with, and is heated in a carbon free furnace, such as air, the carbon will tend to diffuse out of the steel resulting in Decarburization. Carburizing is done by three methods

2. Pack Carburizing:

Parts are packed in a high carbon medium such as carbon (graphite) powder or cast iron shavings and heated in a furnace for 12 to 72 hours at 900 °C (1652 °F). At this temperature carbon mono oxide gas is produced which is a strong reducing agent. The reduction reaction occurs on the surface of the steel releasing Carbon, which is then diffused into the surface due to the high temperature. When enough carbon is absorbed inside the part (based on experience and theoretical calculations based on diffusion theory), the parts are removed and can be subject to the normal hardening methods.

The Carbon on the surface is 0.7% to 1.2% depending on process conditions. The hardness achieved is 60 - 65 RC. The depth of the case ranges from about 0.1 mm (0.004 in) up to 1.5 mm (0.060 in). Some of the problems with pack carburizing is that the process is difficult to control as far as temperature uniformity is concerned, and the heating is inefficient.

Figure -Pack carburizing

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3. Gas Carburizing:

Gas Carburizing is conceptually the same as pack carburizing, except that Carbon Monoxide (CO) gas is supplied to a heated furnace and the reduction reaction of deposition of carbon takes place on the surface of the part. This process overcomes most of the problems of pack carburizing. The temperature diffusion is as good as it can be with a furnace. The only concern is to safely contain the CO gas. A variation of gas carburizing is when alcohol is dripped into the furnace and it volatilizes readily to provide the reducing reaction for the deposition of the carbon.

Figure -Gas carburizing

4. Liquid Carburizing:

The steel parts are immersed in a molten carbon rich bath. In the past, such baths have cyanide (CN) as the main component. However, safety concerns have led to non-toxic baths that achieve the same result.

5. Nitriding

Nitriding is a process of diffusing Nitrogen gas into the surface of steel. The Nitrogen gas forms Nitrides with elements such as Aluminum, Chromium, Molybdenum, and Vanadium. The parts are heat-treated and tempered before nitriding. The parts are then cleaned and heated in a furnace in an atmosphere of dissociated Ammonia (containing N and H) for 10 to 40 hours at 500-625 °C (932 - 1157 °F). Nitrogen diffuses into the steel and forms nitride, and goes to a depth of up to 0.65 mm (0.025 in). The case is very hard and distortion is low. No further heat treatment is required; in fact, further heat treatment can crack the hard case. Since the case is thin, surface grinding is not recommended. This can restrict the use of nitriding to surfaces that require a very smooth finish. At nitriding temperature ammonia gas gets dissociated as follows

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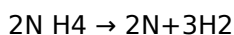
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6. Cyaniding:

Cyaniding is a case hardening process that is fast and efficient; it is mainly used on low carbon steels. The part is heated to 871-954 °C (1600-1750 °F) in a bath of sodium cyanide and then is quenched and rinsed, in water or oil, to remove any residual cyanide.



This process produces a thin, hard shell (between 0.254 - 0.762 mm [0.010 and 0.030 inches]) that is harder than the one produced by carburizing, and can be completed in 20 to 30 minutes compared to several hours so the parts have less opportunity to become distorted. It is typically used on small parts such as bolts, nuts, screws and small gears. The major drawback of cyaniding is that cyanide salts are poisonous.

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